LABORATORY PROJECT No. 1

Financial Fraud Detection – Classification of Highly Imbalanced Data

Project Objective

The objective of the project is to design, implement, and compare machine learning models for detecting fraudulent transactions under conditions of extreme class imbalance (e.g., 0.1–1% fraud cases).

Students are expected to:

- understand the problem of class imbalance,
- analyze the impact of different data balancing strategies,
- compare classical and ensemble models,
- evaluate models using appropriate metrics (not only accuracy),
- conduct a cost analysis of incorrect decisions.

At least one dataset must be selected from Kaggle or the UCI Repository.

Example datasets:

- Credit Card Fraud Detection Dataset (Kaggle) – 284,807 transactions, ~492 fraud cases,
- IEEE-CIS Fraud Detection (more complex, requiring feature engineering).

In real financial systems, the fraud class often constitutes less than 1% of the data. A model may achieve 99% accuracy by classifying all observations as the normal class.

At the same time, the cost of a False Negative error (undetected fraud) is usually significantly higher than the cost of a False Positive.

Therefore, the project should focus on:

- analysis of evaluation metrics under class imbalance conditions,
- strategies for handling class imbalance,
- cost-sensitive classification analysis.

Baseline models: Logistic Regression (baseline), Decision Tree, Random Forest, XGBoost, LightGBM

Class imbalance approaches: No modification (baseline), Class weighting, Random undersampling, Random oversampling, SMOTE, ADASYN, Threshold moving.

Evaluation metrics – It must be demonstrated that: Accuracy ≠ an appropriate metric in fraud detection.

Required metrics: ROC-AUC, PR-AUC (key metric), Recall (fraud), Precision, F1-score, Confusion matrix, Expected cost (cost matrix)

Analyses required in the project:

- The impact of balancing strategies on fraud Recall.
- Decision threshold analysis (threshold tuning).
- Model stability (k-fold cross validation). Comparison of models in terms of effectiveness, training time, stability, and overfitting analysis.

Report 1 – Theoretical aspects and data description

- Introduction to the fraud detection problem.
- Data description.
- Description of models and balancing methods selected for the solution.

Assessment Criteria

Criterion	3.0 (15–17 pts)	3.5 (18–19 pts)	4.0 (20–21 pts)	4.5 (22–23 pts)	5.0 (24–25 pts)
Description of the problem and class imbalance	General description	Correct indication of % fraud	Class distribution analysis	In-depth analysis + visualizations	Cost-sensitive interpretation of the problem
EDA and data description	Basic information	Feature distributions	Correlation analysis	Outlier and missing data analysis	Identification of risks (data leakage)
Justification of model selection	No justification	Brief justification	Proper justification	Comparison of methods	Formulated research hypotheses

Report 2 – Method implementation and evaluation metrics

Implementation code (ZIP file)

Description of the most important implemented functions

Description of evaluation metrics

MANDATORY REQUIREMENT (for report approval)

The student must:

- implement at least 3 classification models,
- apply at least 1 class imbalance handling method,
- compute the following metrics: ROC-AUC, PR-AUC, Recall (fraud).

Failure to include any of the above → report not approved.

Grade	Min. number of models	Technical requirements	Methodological elements	Description
3.0 (21–23 pts)	3	1 balancing method, basic metrics	Train/test split	Correct model implementation. Superficial analysis, no in-depth methodological reflection.
3.5 (24–26 pts)	3	2 balancing methods, ROC and PR curves	Comparison table	Ordered comparison. Visible attempt to analyze differences.
4.0 (27–29 pts)	3–4 (min. 1 boosting)	≥2 balancing methods	k-fold cross validation, variance analysis	Student understands class imbalance and model stability.
4.5 (30–32 pts)	4	≥3 balancing methods	Cross-validation + threshold tuning	Methodological correctness (no data leakage). Precision–recall trade-off analysis. Stable results.
5.0 (33–35 pts)	4+ (min. 2 ensemble)	≥3 balancing methods + cost-sensitive evaluation	Experiment automation, hyperparameter analysis	Very high experimental level. Reproducibility and critical analysis.

Models must include at least: 1 linear model, 1 tree-based model, 1 ensemble model. Balancing methods must be applied exclusively to the training set.

Report 3

RESULTS ANALYSIS AND EXPERIMENTAL CONCLUSIONS (40 pts)

The report must include:

- a consolidated results table for all models,
- analysis of the impact of balancing methods,
- interpretation of results for the fraud class,
- final conclusions.

Analyses required in the project:

- Impact of balancing strategies on fraud Recall.
- Decision threshold analysis (threshold tuning).
- Model stability (k-fold cross validation). Comparison of models in terms of effectiveness, training time, stability, and overfitting analysis.

Mandatory requirement (for report approval)

Lack of analysis (presentation of tables only) → report not approved.

Grade	Scope of analysis	Required elements	Level of interpretation	Description
3.0 (24–26 pts)	Simple model comparison	Results table, confusion matrix	"Model A better than B" type description	Reproductive analysis without deeper interpretation.
3.5 (27–29 pts)	Comparison of several metrics	ROC or PR-AUC analysis	Indication of differences	Attempt at interpretation without causal analysis.
4.0 (30–33 pts)	Balancing impact analysis	Fraud recall, CV results	Precision–recall trade-off interpretation	Understanding of imbalance impact.
4.5 (34–37 pts)	Threshold tuning + stability	Variance analysis, overfitting	Discussion of trade-offs and risks	Mature experimental analysis.
5.0 (38–40 pts)	In-depth cost and practical analysis	Expected cost, trade-off analysis	Interpretation in real financial systems context	Research-level work. Critical evaluation and improvement proposals.

Clarification of requirements for higher grades

From 4.0 and above, the report should include:

- analysis of balancing methods' impact on fraud Recall,
- stability analysis (mean + standard deviation from CV).

From 4.5 and above:

- decision threshold analysis,
- overfitting analysis (train vs validation).

For 5.0:

- cost analysis (Expected cost with cost matrix),
- identification of study limitations (e.g., small number of fraud cases),
- proposals for further system development.

Detailed scoring (40 pts)

1. Consolidated results table (all models + all metrics) – 6
2. Analysis of balancing methods' impact on Recall (fraud) – 8
3. ROC-AUC and PR-AUC analysis (including plots) – 6
4. Stability analysis (k-fold: mean + standard deviation) – 6
5. Decision threshold analysis (threshold tuning) – 5
6. Cost analysis (Expected cost, cost matrix) – 5
7. Quality and maturity of experimental conclusions – 4

Total – 40 pts

Final Laboratory Grade

The final laboratory grade is the sum of points from the three reports according to the following scale:

Total points (R1+R2+R3)	Final laboratory grade
0-59	2
60-67	3
68-74	3,5
75-82	4
83-90	4,5
91-100	5